Clinical Features

Skin

Approximately half of patients with xeroderma pigmentosum (XP) have a history of acute sunburn following minimal UV exposure, while the remainder tan normally without excessive burning. In all patients, numerous freckle-like hyperpigmented macules develop predominantly on sun-exposed skin (see Fig. 140-2). The median age of onset of cutaneous symptoms is between 1 and 2 years (see Fig. 140-3), and these lesions typically spare sun-protected areas such as the buttocks. However, in severely sun-exposed individuals, pigmentary changes may even appear in the axillae.

With continued sun exposure, the skin becomes dry and parchment-like, with increased pigmentation—giving rise to the name *xeroderma pigmentosum* ("dry pigmented skin"; see Fig. 140-2A). Premalignant actinic keratoses develop at an early age (see Fig. 140-2B). The appearance of sun-damaged skin in children with XP resembles that seen in adults with chronic occupational sun exposure, such as farmers or sailors.

Cancer

Patients with XP under the age of 20 have a greater than 1000-fold increased risk of developing cutaneous basal cell carcinoma, squamous cell carcinoma, or melanoma. The median age of onset of skin cancer in XP patients is dramatically reduced by approximately 50 years compared to the general U.S. population (see Fig. 140-3). Internal neoplasms are also increased, with reports suggesting a 10- to 20-fold higher risk.

Eyes

Ocular abnormalities are nearly as common as cutaneous findings and represent a key feature of XP. The posterior eye (retina) is protected from UV radiation by the anterior structures (eyelids, cornea, and conjunctiva), so clinical manifestations are largely confined to these UV-exposed areas.

Photophobia is often present and may be accompanied by prominent conjunctival injection. Schirmer's testing frequently shows reduced tear production, leading to dry eyes. Chronic UV exposure can result in severe keratitis, corneal opacification, and neovascularization.

Eyelids exhibit increased pigmentation, loss of eyelashes, and atrophy, which may lead to ectropion, entropion, or, in severe cases, complete eyelid loss. Benign conjunctival inflammatory masses or papillomas of the eyelids may occur. Malignancies such as epithelioma, squamous cell carcinoma, and melanoma commonly arise in UV-exposed ocular tissues. Ocular manifestations may be more severe in individuals with darker skin.

Neurologic System

Neurologic abnormalities occur in approximately 30% of XP patients. Onset can range from infancy to the second decade of life. Findings vary from mild (e.g., isolated hyporeflexia) to severe, progressive neurodegeneration, including:

- Progressive intellectual disability
- Sensorineural hearing loss (typically starting with high-frequency deficits)
- Spasticity
- Seizures